

Landslide Susceptibility Assessment of Rangamati Hill District Using Deep Learning Method

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Abstract: Landslides are one of the major geological hazards globally, especially in mountainous regions. Considering that 10% of total land occupies hilly terrain, landslide-induced disasters are also prominent in Bangladesh. With rapid urbanization, the development of hilly areas is noticeable, and hence, for safe and economic growth, the susceptible regions need to be identified. Emphasizing the actual instance, this research aims to develop a comprehensive landslide susceptibility map for Rangamati district, one of the important districts of hill tracts of Bangladesh. Both geologic and geotechnical parameters extracted from remote sensing techniques are employed to prepare the initial datasets. Landslide inventory maps from NASA have been used to prepare synthesized datasets for validation. Deep learning that fundamentals on neuron networks of the human brain have been a prospective tool in dealing with highly sparse datasets and random variables. Therefore, a widely used deep learning method, known as Deep Neural Network, has prepared a landslide susceptibility map for the Rangamati hill district. 80% of the data have been used to train the model, while 20% have been used to test the model. The developed susceptibility map portrays an accuracy level of 88% in terms of its statistical distribution. Besides that, the inventory map converges well with the susceptible map; hence, it implicitly substantiates the accuracy of the developed procedure. Overall, this research provides a general view of landslide risk for the Rangamati hill district. With more inventory data, this method can be useful to prepare a zone-wise susceptibility map over the entire hill district.

1. Introduction

Landslide susceptibility mapping helps us discover high-risk areas when historic records are merged with regional topographic data. This complex analysis takes into account a myriad of factors affecting landslides: Landslide studies involve measuring terrain slopes, altitude measurements, lithology, precipitation rates, human land development plus additional physical characteristics of the environment. This mapping strategy becomes essential for several reasons. These large earth movements destroy lives and structures in addition to harming both our natural spaces and human communities. LSM helps identify potential landslide locations which helps with planning land usage and developing better disaster risk management.(Lee et al., 2020) The maps guide which areas need disaster prevention efforts while helping regional planners design land use plans and informing government policy decisions nationwide.

The Hill District in Rangamati, Bangladesh regularly experiences landslides. The area suffers heavy rainfall from monsoons and has steep terrain that regularly produces landslides. The district experiences greater disaster risks because its urban expansion involves heavy hill removal activities. Numerous landslides hit Rangamati and caused fatalities along with massive financial damage to various properties.(Hafsa et al., 2022) The landslide-prone region needs both complete and precise LSM systems to reduce damage from future landslides. Our research develops a Deep Neural Network model to meet its goals. The DNN system can model difficult relationships between environmental conditions and landslide occurrences in the study DNN models handle extensive amounts of data more effectively than other systems. The models detect complicated data patterns across large volumes of information. DNNs produce better results than traditional methods in some research. Studies showed that deep learning models predict landslide susceptibility better than standard benchmark algorithms did.(Habumugisha et al., 2022) These DNN studies reflect earlier LSM research in Rangamati, plus comprehensive academic studies on the topic. Scientists combined remote sensing and GIS techniques to study LSM.(Chowdhury et al., 2024) Studies across Rangamati have shown that landslides depend on several elements like rainfall levels, slope steepness and altitude, soil composition plus forest presence.(Begum et al., 2020) Research outcomes using NDVI as a measurement vary between showing higher landslide rates in less vegetated areas and increased landslide rates in areas with robust vegetation. Research shows mixed evidence about how an area's distance from water bodies and its elevation level together affects landslide risks. Our research uses advanced deep learning through deep neural networks (DNN) to map landslide vulnerability throughout the Rangamati Hill Tract. The research aims to test if the DNN models allow us to produce more reliable landslide predictions than using traditional techniques. This map helps local governments identify specific landslide threats within their areas to create better safety strategies.

2. Study Area

The study area is the Rangamati District which is in southeastern Bangladesh, covering an area of 611613,000 m² (or 6116.13 km²), which lies between 22°27' and 23°44' N latitude; 91°56' and 92°33' E longitude; the Rangamati Municipality Area which is 546.49 km² (Figure 1.a). It is situated on the Arakan Yoma anticlinorium whose strata belong to the Bhuvan formation of the light reddish-brown siltstone and shale characterized by their propensity of landsliding, which is overlaid by the Boka Bill formation of the grayish-brown well-bedded siltstone and shale that is less susceptible to landslide. The district's altitude varies from 50 m to 900 m, with an average altitude of 610 m where 44% of the area lies between 146 m and 576 m. The average annual rainfall at Rangamati is 2,540 mm, of which more than 60% records over 2,500 mm of rainfall under monsoon conditions(Begum et al., 2020). Temperature varies from 36.5 °C to a minimum of 12.5°C. Sometimes, streams, rivers, and lakes, for instance, Kaptai Lake, also play a very important role in the case of landslides. Areas across the southeastern, such as Belaichari and Uraichari, are hilly, with slopes over 30 degrees, which are almost 43% of the whole. The eastern portion exhibits greater geological features than the western side, such as thrusting and folding, due to its tectonic formation.(Hafsa et al., 2022)

2.1 Data Acquisition

In this study, nine factors representing various parameters were employed for the Landslide susceptibility map using previous literature as their guidance. Topographic data were derived from the US Geological Survey STRM Digital Elevation Model (DEM).(Hafsa et al., 2022) The

DEM derived useful parameters like Elevation, Slope, Curvature, and Aspect, the Topographic Wetness Index (TWI), and Stream Power Index (SPI). Elevation data is vital because it affects the potential energy and landslide behavior. Slope values were computed from the DEM, which is an important contributor to flow dynamics and shear stress as the roughness increases with steeper slopes. The DEM was used to extract both plan curvature and profile curvature; plan curvature defines the convergence or divergence of water flow, while profile curvature defines its volume and sediment transport speed. Aspect data that reflect slope direction were also derived from the DEM to examine solar exposure, as this can affect soil moisture and vegetative patterns.(Ahmed, 2015).

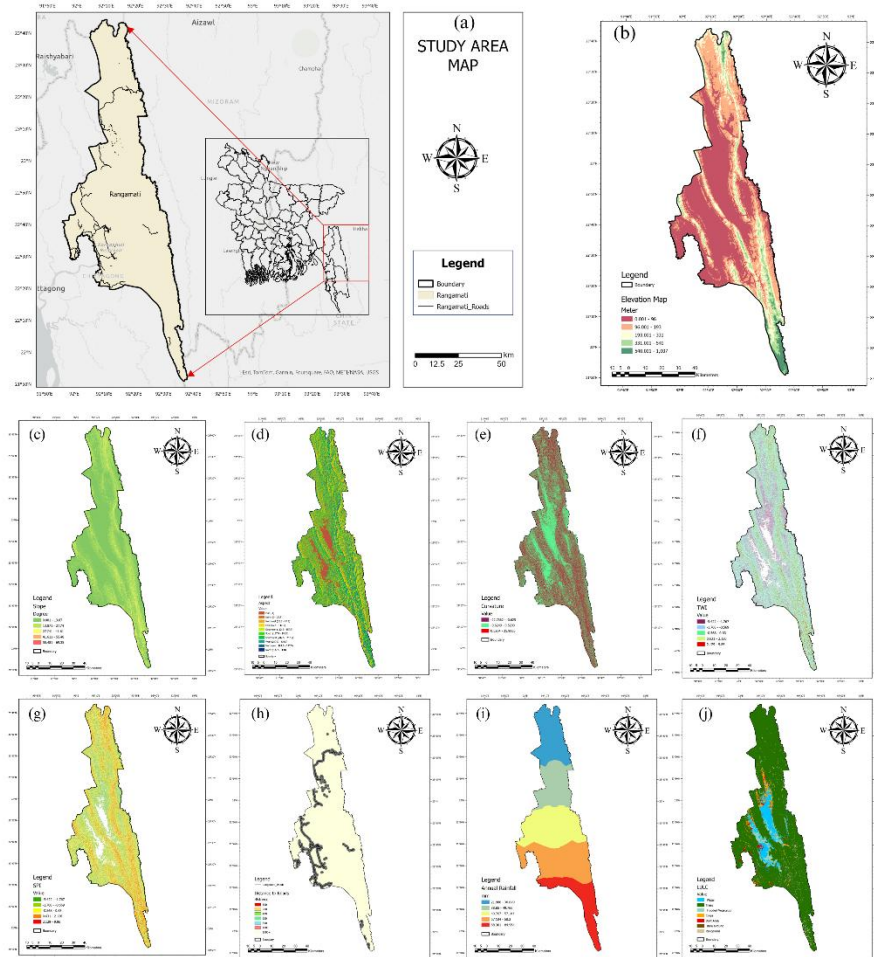


Figure 1. (a) study area map (b) elevation [m], (c) slope [°], (d) aspect, (e) curvature, (f) TWI, (g) SPI, (h) distance to roads [m], (i) Annual Rainfall [mm], and (j) land use land cover [LULC].

The topographic wetness index (TWI) helps to perceive hydrological effects and it is calculated using:

$$TWI = \ln \left(\frac{a}{\tan b} \right) \quad (1)$$

The computational method uses local upslope drainage area values denoted by a and local slope values expressed as $\tan b$. The erosive power of water in motion becomes visible through the analysis of the Stream Power Index (SPI) which is derived from the DEM. It is calculated using:

$$SPI = A_s \cdot \tan\beta \quad (2)$$

The assessment utilizes the catchment area-specific value A_s together with the slope gradient value β . (Habumugisha et al., 2022).

Land Use Land Cover (LULC) data were obtained from ESRI Sentinel-2 imagery and divided into seven classes: water bodies, forests, flooded ruderal vegetation, agricultural land, built-up land, barren land, and grasslands. These classifications help to understand how land use patterns affect landslide susceptibility. Rainfall is a key driver of landslides by increasing soil moisture, elevating pore water pressure, and decreasing soil shear strength for intensity. We also examined distance to roads derived from Geofabrik data, as roads can destabilize slopes through cut-offs and modified drainage pathways. The amalgamation of this varied information — topographical, land cover, climate, and anthropogenic variables — creates a solid base for correctly modeling landslide susceptibility across the area of interest.

2.3 Deep Neural Network (DNN)

Regarding use in LSM, DNN shows high accuracy in capturing complex relationships between geology, topography, and environment. Our Sequential DNN model has two hidden layers, and the layers consist of dense connections, ReLU activation, batch normalization, and the control of dropout to help with regularization. In the processing, forward propagation enables the input data to give the right signals, then neurons are activated through non-linear activation functions. Backpropagation helps modify weights and biases to minimize prediction errors. To overcome overfitting, we use Adam optimizations to fine-tune neural networks' parameters, while early stopping and learning rate reductions are also employed.

3. Results and Discussions

The Deep Neural Network (DNN) model was efficient in predicting landslide susceptibility throughout this research. During testing in parallel training, the deep neural network model achieved 0.8744 accuracy and then 87% on testing data showing that its performance held up across both datasets. This study has obtained an accuracy value of 87% which proves that the adapted model achieved correct classifications for 87% of the dataset samples. The evaluation metrics demonstrate effective model performance through a recall metric of 0.9070, precision of 0.8125, and ROC-AUC score of 0.88 (Figure 2). The landslide susceptibility map generated from the DNN model categorized the study area into five zones: very low, low, medium, high, and very high susceptibility (Figure 3). Analysis of 198 historical landslide occurring points identified that most (190 points) of the locations are in high susceptibility ratings alongside 6 points in high susceptibility zones and an additional 2 points situated in medium susceptibility zones. The observed pattern of landslide data implies additional research because these findings could result from the uneven spatial distribution of landslide records within the study region. The documented point concentration within the high susceptibility zone shows potential correlations with data gathering bias and distinct regional topographical features along with specific triggering events. The performance of the adapted model deviates in different ways when compared to Maoxian County, China's study results which reached a recall of 0.943, an F1 score of 82.8, and an AUC of 0.877.

The differences emerge from topographic changes together with geological conditions and climatic factors while the primary source remains the number and distribution of recorded landslide incidents. The model performance deviation resulting in lower recall scores of 0.9070 for this study as compared to 0.943 in Maoxian County indicates the adapted model did not identify some potential landslide-susceptible regions. (Habumugisha et al., 2022).

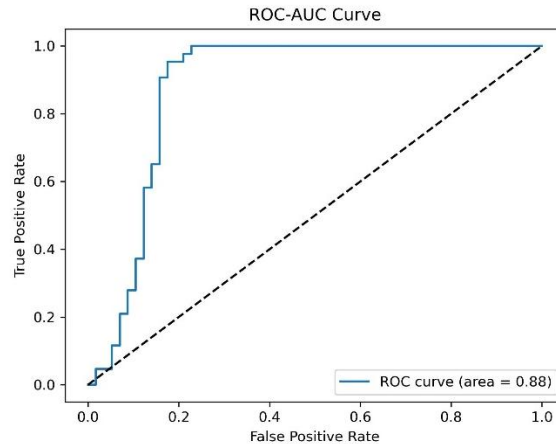


Figure 2: AUC & ROC curve for DNN

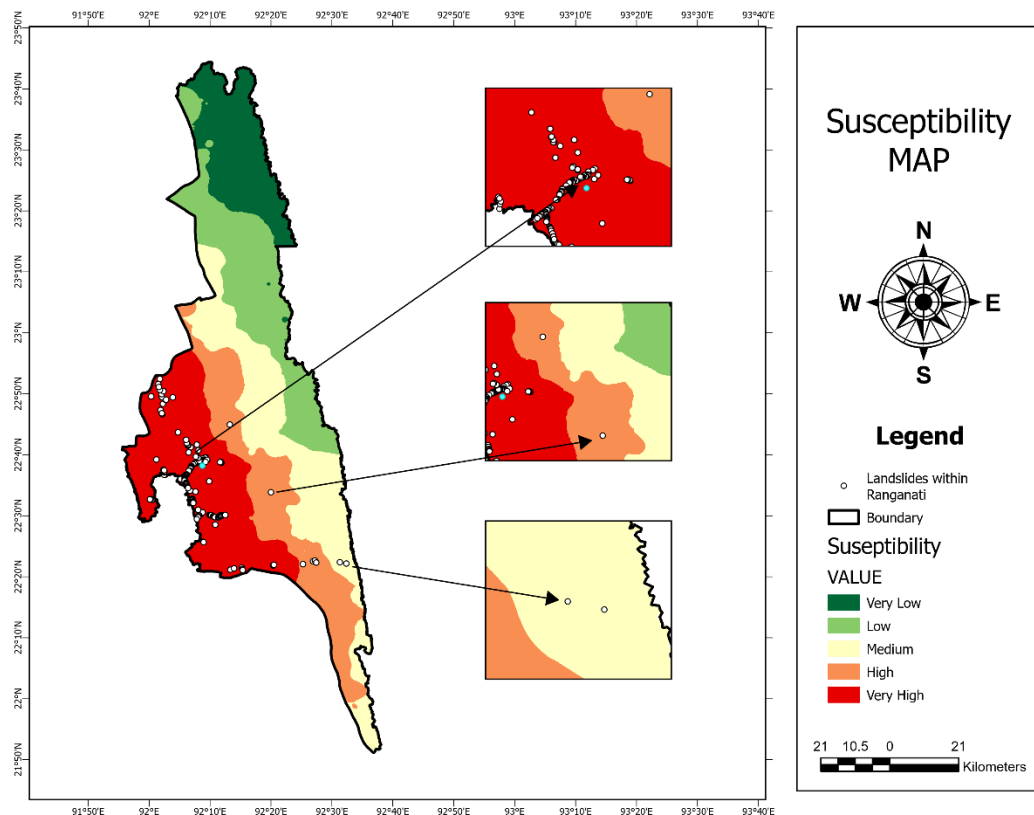


Figure 3: Landslide Susceptibility Map

This study reports a precision measure of 0.8125, showing a moderate false positive rate because of the model's cautious classification system for landslide-prone zones. In landslide risk assessment, professional standards usually favor conservative classification methods because estimating the risk higher tends to prevent underprepared situations.

This study's fundamental restriction lies in the limited inventory of 198 landslide points combined with their uneven distribution across the study area. Better spatial distribution and a greater

quantity of landslide inventory points could lead to higher model accuracy and more precise zoning delineation. A detailed and spatially balanced inventory proves essential for successful landslide susceptibility mapping.

The landslide susceptibility mapping task shows the DNN model as highly effective according to study performance results. Effective discrimination between varying landslide susceptibility regions is demonstrated by the model through its high values in accuracy alongside recall and ROC-AUC scores. The study findings emphasize that we require spatially comprehensive and resilient landslide inventories for better analysis results.

4. Conclusions

Using the Deep Neural Network (DNN) model, this research successfully developed a landslide susceptibility map for the Rangamati Hill District located in Bangladesh. Results from the model showed a good accuracy of 88%, demonstrating the capacity to predict areas prone to landslides. The resulting susceptibility map divides the region into five zones based on low, low, medium, high, and very high susceptibility—providing valuable insights for local authorities and disaster management agencies. The results show that most historical landslide events are in high-susceptibility areas, highlighting the urgent demand for a sophisticated control strategy in this region. However, the study's limitations, specifically in the poor distribution of landslide inventory data, may have impacted the model's predictive ability. An area of future research is to expand the dataset to include more inventory points to improve model accuracy and reliability. This study generally helps to understand landslide risks in hilly terrain and shows how the potential deep learning techniques can be utilized for environmental hazard assessments. The findings can aid in informed policy formation for urban planning and disaster preparedness, ultimately contributing to safer development practices in vulnerable regions.

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